SEHS2042 Computer Programming
Group Project – Virtual Library Management System
(Due: 23:59, 7 December 2026, Monday)

Expected Learning Outcomes

- develop computer programs in one or more high level language programming environment;
- design and develop structured and documented computer programs;
- explain the fundamentals of object-oriented programming and apply it in computer program development;
- integrate computer programming techniques to solve practical problems.

Introduction

In this project, you are going to form group with **5 to 6 members** and develop a "Virtual Library Management System" that runs in the command line environment. The system manages library reader accounts and their reading activities. The system allows library staff to check account details, create borrowing records, calculate reading points, and handle point transfer between readers.

Tasks

- Each group is required to write a Win32 Console Application program called **VLM.cpp**.
- Each student is required to submit a Peer-to-Peer evaluation form (through the given Word file) via Blackboard.

Program Requirements

R0 When the program starts, the console should display a welcome message, followed by the Main Menu of the program. Users can enter the options of the corresponding actions (see **R1** to **R6** below).

```
Welcome Message designed by your group
*** VLM Main Menu ***
[1] Load Starting Data
[2] Show All Reader Accounts
[3] Open or Close Reader Account
[4] Reader Account Operations
[5] Generate Borrowing and Transaction Statement
[6] Credits and Exit
************************************************
Option (1 - 6):
```

R1 [1] Load Starting Data

When the user inputs 1 in the Main Menu, the system is loaded with starting data. The starting data includes reader account records as shown in Table 1, the borrowing records in Table 2, and the system date setup in R1.2. After the starting data is loaded, the system returns to the Main Menu.

R1.1 The starting data of reader account records to be loaded, together with the required data format of the data fields, are described below.

Reader Account records

Reader ID	Library Card Number	Reader Name	Fiction Rollover Balance	Non-Fiction Rollover Balance	Reading Points Balance
202456734	8066778JW14	WONG Claire	0	0	100
202333890	8378678JM15	MA Kathy	0	0	200
202067856	8138876JC24	CHAN Peter	0	0	500
202211843	8689000JC11	CHEUNG Alice	0	0	1000

Table 1. Reader Account Records

Borrowing records

Reader ID	eBook ID	Borrowing Date	Due Date	Updated
202211843	F01			0
202211843	F02			0
202333890	F01			0
202067856	N05			0
202067856	N02			0
202211843	N01			0

Table 2. Borrowing Records

Data format of each field:

- **Reader ID**: A string that uniquely identifies a reader account in the system. The first four digits indicate the year the reader account was created, followed by five random digits.

- **Library Card Number:** A string that must be unique within the system. The format complies with the guidelines set by the Joint University Librarians Advisory Committee (JULAC).
    - Format: 2-digit Institutional ID + 5-digit Sequential Number + a static letter "J" for JULAC + Reserved Letter + Card Replacement Digit (1-9) + Check Digit (0-9).

    - Institutional ID:
        - 80: The University of Hong Kong (HKU)
        - 81: The Chinese University of Hong Kong (CUHK)
        - 82: The Hong Kong University of Science and Technology (HKUST)
        - 83: The Hong Kong Polytechnic University (PolyU)
        - 84: City University of Hong Kong (CityU)
        - 85: Hong Kong Baptist University (HKBU)
        - 86: Lingnan University (LU)
        - 87: The Education University of Hong Kong (EdUHK)

    - 5-digit Sequential Number: It is a 5-digit unique number from 10000 to 99999 assigned to the cardholder and should be entered by users.

    - Reserved Letter: It is first letter of reader's surname and should not be entered by users.

    - Card Replacement Digit: It should not be entered by users.
        - First Card: The very first library card issued to a reader has a replacement digit of 1.
        - Replacement Card: If that card is lost and replaced, the replacement digit increments. For example, the new card would have a replacement digit of 2.
        - Subsequent Replacements: If that card is lost again, the next card would have a digit of 3, and so on, up to a maximum of 9.

    - Check Digit: The system should generate this check digit automatically when opening a new reader account, and it should not be entered by users. For instance, the check digit for library card number 8012345JA13 is calculated as follows:
        - Character Values: Convert the character to numerical values (A=10, B=11, ..., Z=35)
        - Position Values: Assign weights to each position from left to right: 7, 3, 1, 7, 3, 1, … repeating, then multiple each numeric value by its weight.

- 8: 8 x 7 = 56
- 0: 0 × 3 = 0
- 1: 1 × 1 = 1
- 2: 2 × 7 = 14
- 3: 3 × 3 = 9
- 4: 4 × 1 = 4
- 5: 5 × 7 = 35
- J: 19 x 3 = 57
- A: 10 × 1 = 10
- 1: 1 × 7 = 7

Total = 56 + 0 + 1 + 14 + 9 + 4 + 35 + 57 + 10 + 7 = 193

Check digit = Total mod 10 = 193 mod 10 = 3

- **Reader Name**: A string that consists of surname and given name. Surnames should be formatted to all capital letters. The full name is assumed to be at most 30 characters long, including white spaces.

- **Fiction Rollover Balance:** The number of Fiction books that have not yet been paired into a Genre Set. It is an integer that ranges from 0 to 3 (see R4.2 below).

- **Non-Fiction Rollover Balance:** The number of Non-Fiction books that have not yet been paired into a Genre Set. It is an integer that ranges from 0 to 3 (see R4.2 below).

- **Reading Points Balance:** An integer number that represents the total number of reading points accumulated in a reader account.

- **eBook ID:** A string that consists of two components.
  - 1 uppercase letter (either F or N) + 2 digits (00-99)

- **Borrowing Date:** A string in the format of YYYY-MM-DD.

- **Due Date:** A string in the format of YYYY-MM-DD. The due date is 21 days after the borrowing date.

- **Updated:** A Boolean value that indicates whether the reading points balance in Reader Account Records has been updated.

R1.2 Immediately after loading the starting data, the system must prompt the user to set the system date.

```
Starting data loaded successfully!

*** Set System Date ***
Please enter system date (YYYY-MM-DD):
```

The date must be a string in format "YYYY-MM-DD" and must fall within the range of 2026-01-01 and 2026-12-31 (inclusive). When the user provides a valid input, the borrowing date in the borrowing records is set to the system date, the due date is updated accordingly, and the system displays a confirmation message. If the user enters an invalid format or a date outside the valid range, the system displays an appropriate error message and prompts the user to try again.

However, if the user enters an invalid date three consecutive times, the system will automatically set the borrowing date to the default date of "2026-06-30", the due date is updated accordingly, and the system displays a warning message, and returns to the Main Menu.

R1.3 Options 2 to 5 on the Main Menu are enabled only after the system is loaded with the starting data. If the user enters options 2 to 5 before starting data is loaded, an error message should be shown, and then the system returns to the Main Menu.

R2 [2] Show All Reader Accounts

[After the starting data is loaded] When the user inputs 2 in the Main Menu, the system displays all the fields of reader records as shown in Table 1 and 2. Tabular format, in two tables, should be displayed for both reader account records and borrowing records. For reader account records (Table 1), all records are displayed in ascending alphabetical order based on the Reader Name; for borrowing records (Table 2), records are displayed in descending order based on Borrowing Date, excluding those with Updated set to true (i.e., 1).

Data shown should be the **latest** set of data since the starting data is loaded (updates on the reader records resulted by the operations under options 3 or 4 in the Main Menu should be included). After showing the data, the system returns to the Main Menu.

R3 [3] Open or Close Reader Account

[After the starting data is loaded] When the user inputs 3 in the Main Menu, the system prompts the user to enter a Reader ID. If the entered Reader ID does not exist in the system, it is an *open reader account* operation. Otherwise, it is a *close reader account* operation.

You can assume the system will store at most 100 reader account records.

Open Reader Account
The system asks the user to input the required information of a new reader: (1) Reader Name (2) Institutional Number and (3) Sequential Number. After getting all user inputs, the system should generate Reader ID and Library Card Number, and then the new reader should be added into the system with the following initial values:

- Reading Points Balance: 0
- Fiction Rollover Balance: 0
- Non-Fiction Rollover Balance: 0

For any invalid input (e.g., invalid Sequential Number), the system allows **TWO** more retries. With more than **THREE** times of invalid inputs, no reader is added. The system prints an appropriate error message and returns to the Main Menu.

Close Reader Account
The system displays the information of the reader (including all the fields in Table 1), and prompts for user's "Yes/No" confirmation on the delete operation. The reader record and corresponding borrowing records are deleted from the system if it is confirmed.

A message showing the summary of the above operation is then displayed, and the system returns to the Main Menu.

R4 [4] Reader Account Operations

[After the starting data is loaded] When the user inputs 4 in the Main Menu, the system prompts for the user input of a Reader ID. If such Reader ID does not exist in the system, the system displays an error message and returns to the Main Menu. Otherwise, it displays the Reader Account Operations Menu as shown below and allows further processing on the reader (e.g., CHEUNG Alice) as follows (see R4.1 to R4.5 below).

```
Action for Reader ID: 202211843
***** Reader Account Operations Menu ***********************
[1] Edit Reader Information
[2] Update Reading Points Balance based on Borrowing Records
[3] Create Borrowing Records
[4] Redeem Reading Points for a Gift & Transfer
[5] Replace Library Card
[6] Return to Main Menu
*********************************************************
Option (1 - 6):
```

R4.1 [1] Edit Reader Information

When such option is chosen, the system shows the current reader information and prompts for the user input of Reader Name, Institutional ID, and Sequential Number. Then the system prompts for user's "Yes/No" confirmation. The reader's information, including Library Card Number, is updated if it is confirmed. The system shows the changes and then returns to the Reader Account Operations Menu.

For any invalid input (e.g., invalid Sequential Number), the system allows **TWO** more retries. With more than **THREE** times of invalid inputs, no change is made. The system prints an appropriate error message and returns to the Reader Account Operations Menu.

R4.2 [2] Update Reading Points Balance according to Borrowing Records

When such option is chosen, the system will calculate and display the reading points balance, and update borrowing records, with the following steps.

- The system first displays all borrowing records for the reader that have not yet had their reading points balance updated. For example, if the system date is set to June 30, 2026, and the reader is CHEUNG Alice (Reader ID: 202211843), the system will display her outstanding borrowing records as shown below.

Reader ID	eBook ID	Borrowing Date	Due Date	Updated
202211843	F01	2026-06-30	2026-07-21	0
202211843	F02	2026-06-30	2026-07-21	0
202211843	N01	2026-06-30	2026-07-21	0

- For the borrowing records that have not had their reading points balance updated, the system calculates reading points earned based on the rules below.

    - Rule 1: Borrowing Bonus
        - Readers earn 50 reading points for every unique book they borrow.
        - If a reader borrows the same book multiple times, it counts only once for points.
        - This applies to all books, regardless of genre, length, or format
        - Example: Reader borrows C++, Python, and C++ again → He/she earns points for 2 unique books (C++ and Python) = 100 points

    - Rule 2: Genre Explorer Bonus
        - Readers earn 100 reading points for every complete "Genre Set"
        - A Genre Set = 1 Fiction book + 1 Non-Fiction book
        - This bonus repeats indefinitely - every time a reader completes a Set, they earn another 100 points
        - Example:
            - Reader has borrowed 3 Fiction books and 2 Non-Fiction books.
            - The system pairs them into 2 complete sets (2 Fiction + 2 Non-Fiction)
            - The remaining 1 Fiction book will roll over and count toward his/her next Set (hint: using Fiction/Non-Fiction Book Rollover Balance).

    - Rule 3: Order Doesn't Matter
        - The system looks at the reader's total borrowing record, not the sequence in which they borrowed the books
        - Fiction and Non-Fiction books can be borrowed in any order
        - Example:
            - Reader borrows: Non-Fiction, Non-Fiction, Fiction, Fiction.
            - The system still counts: 2 Fiction and 2 Non-Fiction = 2 complete Sets.
            - He/she earns: 2 × 100 = 200 reading points

    - Rule 4: No Duplicate Counting for Bonuses
        - Only unique book titles count toward completing a Genre Set.
        - Borrowing the same book multiple times does not help complete additional Sets
        - Example:
            - Reader borrows: Legend (Fiction), Legend (Fiction again), C++ (Non-Fiction)

        - The system counts: 1 unique Fiction and 1 unique Non-Fiction = 1 Set
        - The duplicated Legend does not count again

    - Rule 5: Rollover Balance Cap
        - The Fiction Rollover Balance and Non-Fiction Rollover Balance cannot exceed 3 each.
        - Non-zero Rollover Balance can count towards future Genre Set.
        - After all Genre Sets have been formed, any remaining unpaired Fiction or Non-Fiction books are added to the corresponding rollover balance.
        - If the resulting balance exceeds 3, it is capped at 3. Any excess books beyond the cap are not carried forward (they are effectively discarded and do not count toward future Genre Sets).
        - Example:
            - Suppose a reader has:
              Fiction Rollover Balance: 3
              Non-Fiction Rollover Balance: 0
            - Reader borrows: Legend (Fiction)
            - Reading points earned: 50 (Borrowing Bonus) + 0 (Genre Explorer Bonus). Fiction Rollover Balance and Non-Fiction Rollover Balance: unchanged.

- The system calculates reading points earned. For instance, CHEUNG Alice (Reader ID: 202211843) will earn a total of 250 reading points:
    - Borrowing Bonus
        - eBook ID: F01, Borrowing Date: 2026-06-30 → 50 points
        - eBook ID: F02, Borrowing Date: 2026-06-30 → 50 points
        - eBook ID: N01, Borrowing Date: 2026-06-30 → 50 points
    - Genre Explorer Bonus
        - Fiction, Fiction, Non-Fiction → 100 points (1 Genre Set)

- Subsequently, the system updates reading points balance and fiction/non-fiction rollover balance in reader account record, and status in borrowing records.
    - In reader account records, reading points balance should be updated to 1250 and fiction rollover balance should be updated to 1.
    - In borrowing records, the Boolean value (Updated) of the following records should be set to true.
        - eBook ID: F01, Borrowing Date: 2026-06-30
        - eBook ID: F02, Borrowing Date: 2026-06-30
        - eBook ID: N01, Borrowing Date: 2026-06-30

- Finally, the system displays the total reading points earned, reading points calculation breakdown, and the updated reader account records, and returns to the Reader Account Operations Menu.

R4.3 [3] Create Borrowing Records

When such option is chosen, the system prompts the user to enter eBook ID, and display genre and title accordingly. In the system, eBooks are defined and remain unchanged according to Table 3 below.

eBook ID	Genre	Title
F01	Fiction	Legend
F02	Fiction	The hunger games
F03	Fiction	Punching the air
F04	Fiction	Looking for Alaska
F05	Fiction	The big sleep
N01	Non-Fiction	C++
N02	Non-Fiction	Python
N03	Non-Fiction	Java
N04	Non-Fiction	HTML5
N05	Non-Fiction	PHP

Table 3. eBook Table

The system then prompts for user's "Yes/No" confirmation, and a new borrowing entry is added into Table 2 if it is confirmed. The borrowing date is set to the system date, and the due date is updated accordingly. The system shows the changes and then returns to the Reader Account Operations Menu.

R4.4 [4] Redeem Reading Points for a Gift & Transfer

When such option is chosen, the system provides two sub-options: Redeem Reading Points for a Gift or Transfer.

(1) Redeem Reading Points for a Gift: The system prompts the user to select a gift number from Table 4. The reading points required for the chosen gift must not exceed the current reading points balance. Upon confirmation, the balance is updated accordingly.

Gift Number	Description	Reading Points Required
1	Movie voucher	200
2	$100 Bookstore voucher	500
3	Library membership (1 year)	1000

Table 4: Redemption Catalog

(2) Transfer: The system prompts the user to enter the target reader account number and the number of reading points to transfer. The system then performs the following checks:
    - the target reader account must exist in the system;
    - the reading point to transfer must be positive and not exceed the source reader account balance;
    - the system then shows the target reader name and asks for confirmation, to avoid wrong transfers.
    - if the user confirms the information, the transfer is executed by deducting the reading points from the source reader's balance and adding them to the target reader's balance.
    - for any invalid input (e.g., insufficient balance, wrong target reader account), the system displays an error and aborts the operation.

The system shows the changes and then returns to the Reader Account Operation Menu.

R4.5 [5] Replace Library Card

When such option is chosen, the system then prompts for user's "Yes/No" confirmation, and the replacement digit increments and check digit is updated accordingly if it is confirmed.

R4.6 [6] Return to Main Menu

When such option is chosen, the system first displays the **latest** reader account record and all borrowing records that have not had their reading points balance updated and then returns to the Main Menu.

R5 [5] Generate Borrowing and Transaction Statement

[After the starting data is loaded] When the user inputs 5 in the Main Menu, the system prompts for the user input of a Reader ID. If such Reader ID does not exist in the system, the system displays an error message and returns to the Main Menu. Otherwise, it displays the transactions, borrowing history as shown below, with reader account information and summary.

```
Reader Name: CHAN Peter
Reader ID: 202067856
Library Card Number: 8138876JC24
Statement Date: 30-06-2026
```

```
------------------------------------------------------------
Transactions:
Type              Reading Points    Description
Redemption        -500              Gift #2: $100 Bookstore voucher

------------------------------------------------------------
Borrowing History:
eBook ID    Title             Borrowing   Due
                              Date        Date
N05         PHP               2026-06-30  2026-07-21
N02         Python            2026-06-30  2026-07-21
------------------------------------------------------------
Reader Account Summary:
Total Reading Points Balance                    : 500
Fiction Rollover Balance                        : 0
Non-Fiction Rollover Balance                    : 0
```

After displaying the Borrowing and Transaction statement, the system returns to the Main Menu.

R6 **[6] Credits and Exit**

When the user inputs this option, the system prompts for user's "Yes/No" confirmation. If the user confirms, the system displays the personal particulars (student name, student number, tutorial group) of the group members and terminates. Otherwise, the system will return to the Main Menu.

R7 For all Yes/No confirmation, the user should input 'Y' or 'y' for Yes and 'N' or 'n' for No. Other input is not acceptable, and the system should ask the user to confirm again.

R8 Suitable checking on user's input is expected, except in situations with assumptions stated in the requirements above. Appropriate error messages should be printed whenever unexpected situations happen, e.g., input value out of range, incorrect character input, etc.

R9 The use of functions (in addition to main function) and classes (i.e., OOP design) are expected in your program. Appropriate comments should be added in your source code file.

R10 Creativity and Critical Thinking: Use suitable format to present all required information of reader account, borrowing records and library operations clearly and neatly. Additional features can be added.

Tips

1. To handle unexpected input error (e.g. input a character to an integer variable), you may use the following code appropriately in your program:

```
cin.ignore();    // Discard the content in the input sequence.
cin.clear();     // Reset the input error status to no error.
```

Submission

Source Code File: Each group submits one source code file (i.e., **VLM.cpp**).
Peer-to-peer Evaluation: Each student fills in the peer-to-peer evaluation form (MS Word) and submits via Blackboard.

All submission should be done **through Blackboard by 23:59, 7 December 2026 (Monday)**. Late submission is subject to 20% deduction in your final marks for each day. No late submission is allowed **4** days after the due date.

Components Weighting

Marks for program (VLM.cpp) are determined by the group-based marks (80%) and percentage of individual contribution (20%), where the percentage of individual contribution is directly proportion to the average marks given by group members in the peer-to-peer evaluation.

Grading Criteria (Program)

Your program will be executed with different test cases in **Microsoft Visual Studio**. Any deviation from the requirement is considered incorrect and **no mark** is given for that case. Your program will also be marked based on its user-friendliness and creativity (e.g., information display, appropriate prompt messages and action result messages if needed).

Aspects	**Percentage**
Program correctness (Follow ALL instructions, marks deduction on errors found)	70%
Program design (Appropriate use of functions, use of class, modularity, etc.)	5%
Program standard (Use of variable names, indentation, line spacing, clarity, comments, etc.)	5%

Algorithm design (Use of reasonable algorithms and data structures)	5%
User-friendliness (Clear guidelines to users, messages to users, etc.)	5%
Creativity and critical thinking (Additional useful features)	10%
Total (Group Mark)	**100% (max)**

Note: the length of your program does not affect the grading of the assignment. However, appropriate use of loops and functions are expected to avoid too many repeated codes in your program, which contributes to the program design score of this assignment.

Marks Deduction

Syntax error: 100% deduction. You will get **0 mark** if your program fails to be compiled.
Runtime error: No mark for the particular test case that triggers the runtime error.
Logic error (bug): No mark for the particular test case that deviates from the requirement. Note that a logic error may lead to **failure in ALL test cases** of one particular set of requirements.

***** Ensure the originality of your work. Plagiarism in any form is highly prohibited. *****

- End –